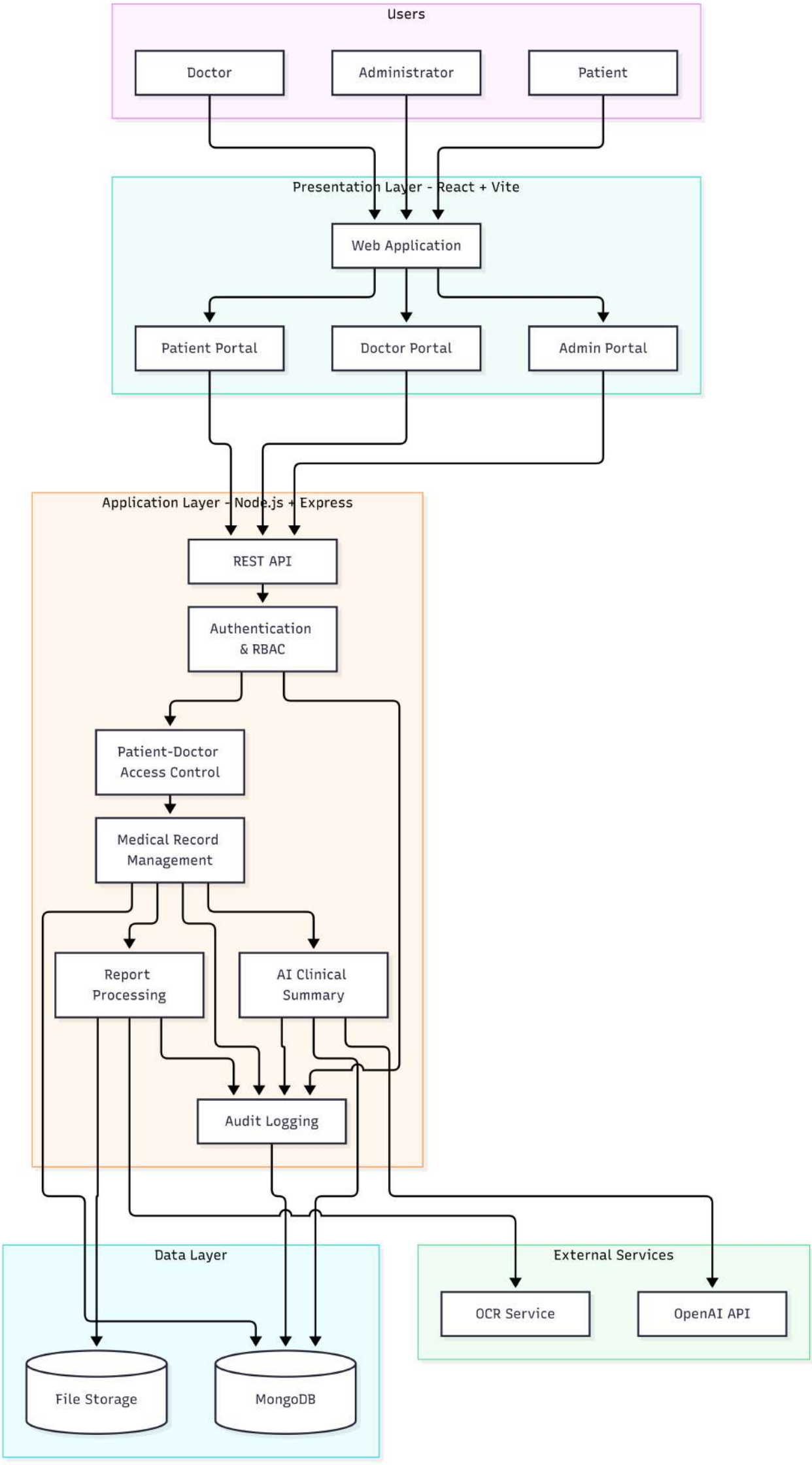
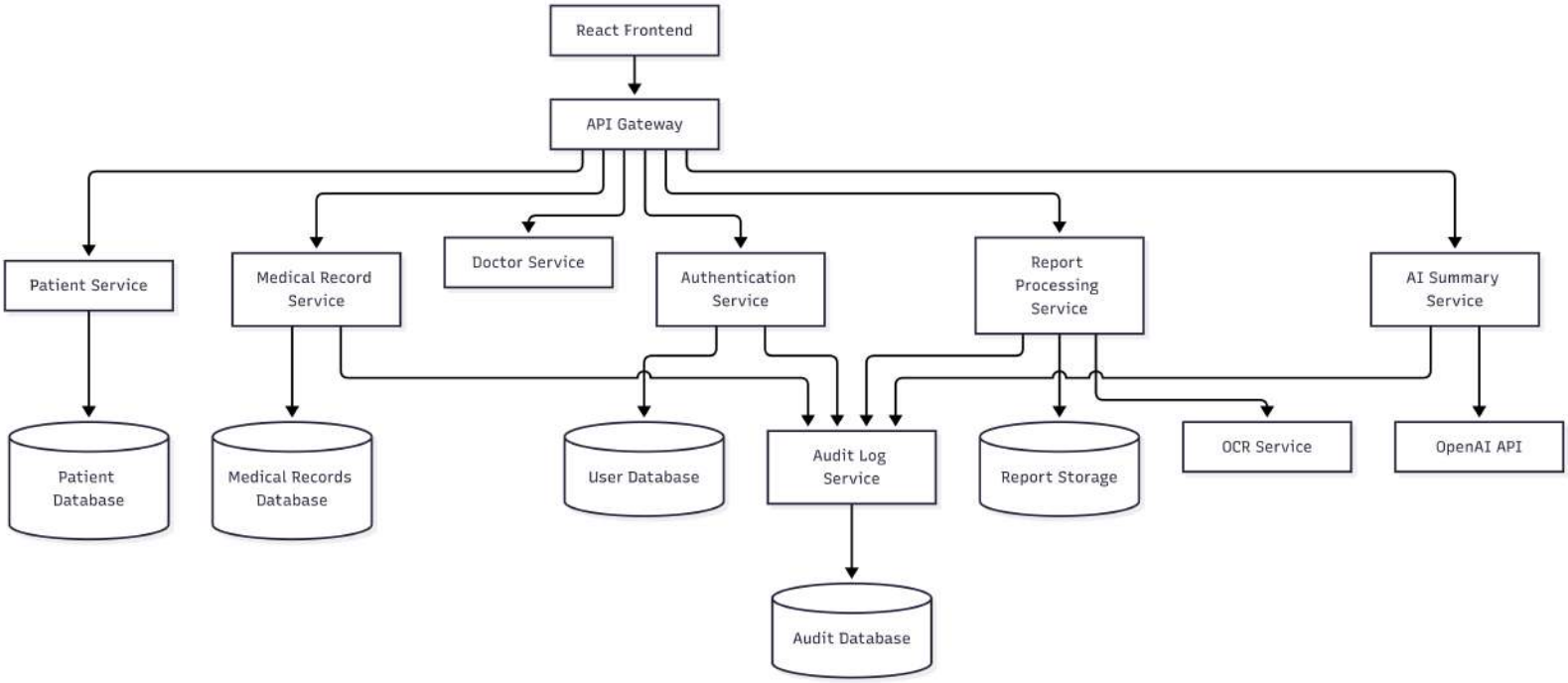
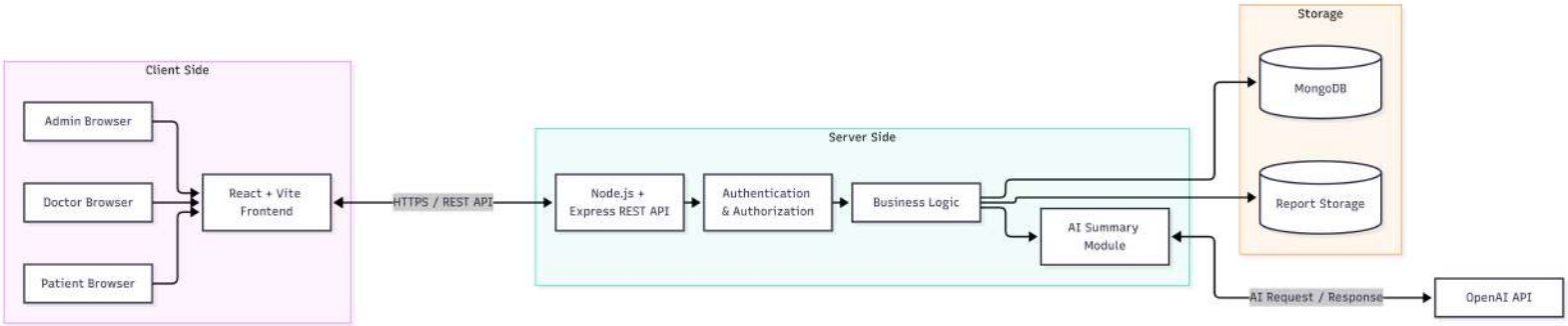
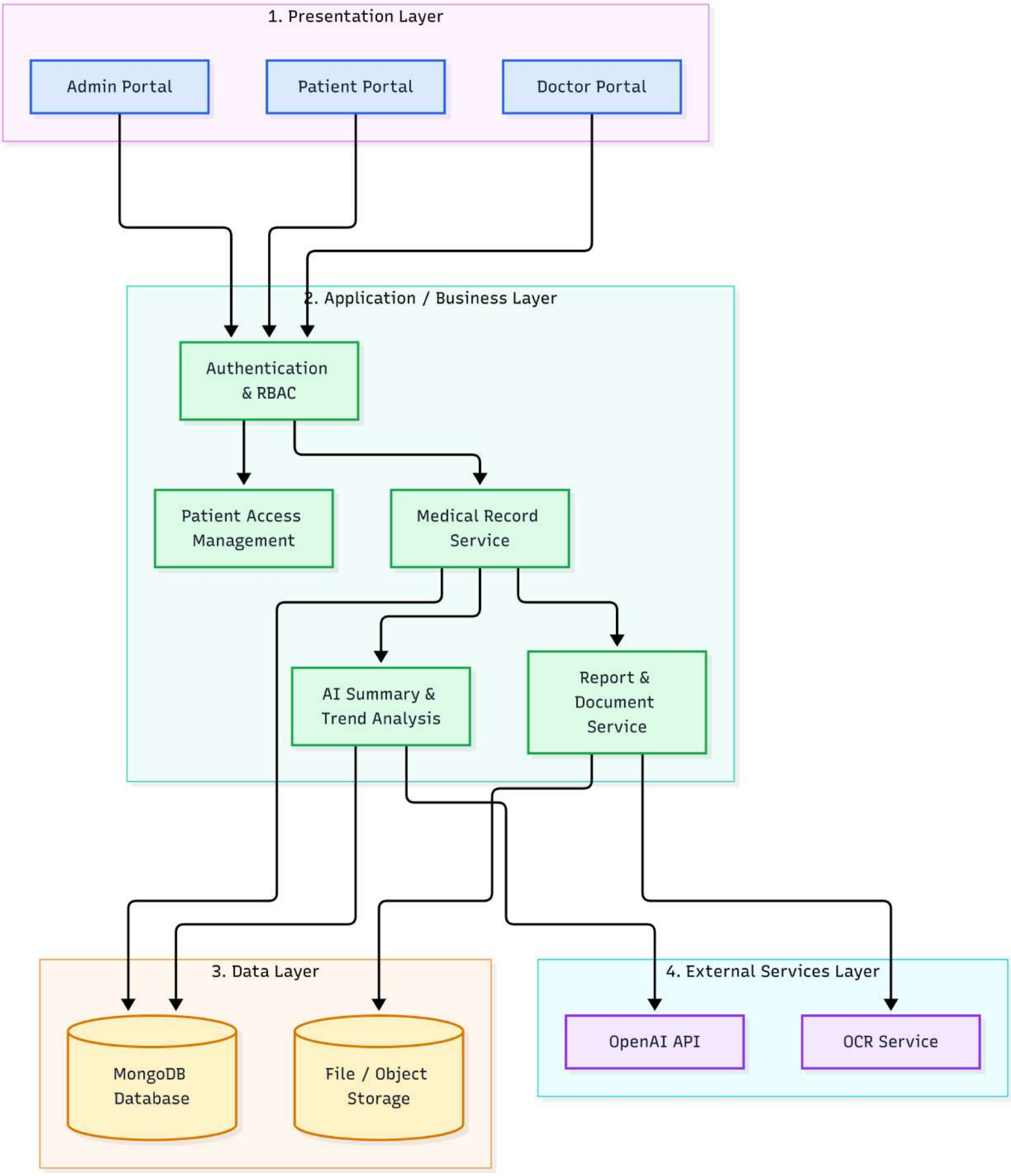

Users
Doctor
Administrator
Patient
Presentation Layer - React + Vite
Web Application
Patient Portal
Doctor Portal
Admin Portal
Application Layer - Node.js + Express
REST API
Authentication & RBAC
Patient-Doctor Access Control
Medical Record Management
Report Processing
AI Clinical Summary
Audit Logging
Data Layer
File Storage
MongoDB
External Services
OCR Service
OpenAI API


React Frontend
API Gateway
Patient Service
Medical Record Service
Doctor Service
Authentication Service
Report Processing Service
AI Summary Service
Patient Database
Medical Records Database
User Database
Audit Log Service
Report Storage
OCR Service
OpenAI API
Audit Database


Client Side
Admin Browser
Doctor Browser
Patient Browser
React + Vite Frontend
HTTPS / REST API
Server Side
Node.js + Express REST API
Authentication & Authorization
Business Logic
AI Summary Module
Storage
MongoDB
Report Storage
AI Request / Response
OpenAI API


1. Presentation Layer
Admin Portal
Patient Portal
Doctor Portal
2. Application / Business Layer
Authentication & RBAC
Patient Access Management
Medical Record Service
AI Summary & Trend Analysis
Report & Document Service
3. Data Layer
MongoDB Database
File / Object Storage
4. External Services Layer
OpenAI API
OCR Service